CSE2331

Graph Algorithms Homework

Problem 1. Consider the following function whose input is an undirected graph G.

```
function Func(G)
    s, r, t = 0
    foreach vertex v_i ∈ G.V do
        foreach edge (v_i, v_j) ∈ G.E incident on v_i do
            foreach vertex v_k ∈ G.V do=
                s = s + 2
            end for
            r = r + 3
        end for
        t = t + 4
    end for
    return st + r
end function
```

Let n be the number of vertices and m be the number of edges of G. Assume the edges of G are represented by an adjacency list.

(a) Analyze the running time of the function and determine its complexity in terms of n and m.

(b) When the code finishes executing, what value is returned by Func?

a) $T(n,m) = \sum_{i=1}^{n}\left[c + \sum_{j=1}^{\deg(v_i)} \sum_{k=1}^{n} c\right]$

$= \sum_{i=1}^{n}\left[c + cn\deg(v_i)\right]$

$= cn + cn\sum_{i=1}^{n}\deg(v_i)$

$= cn + cn(2m)$

So, $T(n,m) \in \Theta(n + nm)$

Recall:

$\sum_{i=1}^{n}\deg(v_i) = 2m$

b) $s = \sum_{i=1}^{n}\sum_{j=1}^{\deg(v_i)}\sum_{k=1}^{n} 2 = 4nm$

$r = \sum_{i=1}^{n}\sum_{j=1}^{\deg(v_i)} 3 = 6m$

$t = 4n$

Returns $st + r = 4nm(4n) + 6m$

Problem 2. Consider the following function whose input is an undirected graph G. Edges of G are represented by an adjacency LIST and that weight(v_i, v_j) is a positive edge weight of edge (v_i, v_j) and computing weight(v_i, v_j) takes constant time. Assume that:

1. `Q.Insert`(x) inserts x into Q and takes $\Theta(\log(s))$ time,
2. `Q.ExtractMin`$()$ removes and returns the minimum element from Q and takes $\Theta(s)$ time.

where s is the number of elements currently in Q.

```
function Func(G)
    Q.Init()
    foreach vertex v_i ∈ G.V do
        foreach vertex v_j in G.V do
            foreach edge (v_j, v_k) ∈ G.E incident on v_j do
                w = weight(v_i, v_j, v_k)
                Q.Insert(w)
            end for
        end for
    end for
    foreach vertex v_i ∈ G.V do
        foreach edge (v_i, v_j) ∈ G.E incident on v_i do
            x = Q.ExtractMin()
            Print x
        end for
    end for
end function
```

$s=1$

$s=2nm$

$s=2nm-2m$

$$\#\text{ inserts} = \sum_{i=1}^{n}\sum_{j=1}^{n}\sum_{k=1}^{\deg(v_j)} 1 = 2nm$$

$$\#\text{ extracts} = \sum_{i=1}^{n}\sum_{j=1}^{\deg(v_i)} 1 = 2m$$

Let n be the number of vertices and m be the number of edges of G. Analyze the run time of the function and determine its complexity in terms of n and m. Assume $m \geq n$.

$$T(n,m) = \sum_{s=1}^{2nm} c\log(s) + \sum_{s=2nm-2m}^{2nm} cs$$

UB: $T(n,m) \leq \sum_{s=1}^{2nm} c\log(2nm) + \sum_{s=2nm-2m}^{2nm} c(2nm)$

$T(n,m) \leq 2nm \cdot c\log(2nm) + \underbrace{(2nm-(2nm-2m)+1)}_{2m+1} c(2nm)$

$O(nm^2)$

LB: $T(n,m) \geq \sum_{s=2nm-2m}^{2nm} cs$

$T(n,m) \geq \sum_{s=2nm-2m}^{2nm} c(2nm-2m)$

$T(n,m) \geq (2m+1)c(2m)(2n-1)$

$\Omega(m^2n)$

Thus, $T(n,m) \in \Theta(nm^2)$

Problem 3 (Optional). Consider the following algorithm,

```
function Original(G)
    Q.Init()
    foreach vertex v_i ∈ G.V do
        Q.Insert(v_i, w(v_i))
    end for
    Analyze(G, Q)
end function
```

$\sum_{s=1}^{n} \log(s) \in \Theta(n\log(n))$

$s = n$

$\Theta(m + n\log(n))$

```
function Analyze(G, Q)
    x = 0
    if Q.Size() ≠ 0 then
        (v_j, w) = Q.ExtractMax()
        x = x + w
        foreach edge (v_j, v_k) ∈ G.E incident on v_j do
            x = x + Weight(v_j, v_k)
        end for
        x = x + Analyze(G, Q)
    end if
    return x
end function
```

$\log(s)$

$\deg(v_j)$

$T(s-1)$

Assume that Q is a priority queue implemented as a Max-Heap in the same manner as the notes. Further assume that w and `Weight` both evaluate to integers and take constant time to compute.

Determine the asymptotic running time of `Original` on a graph G, where $|G.V| = n$ and $|G.E| = m$.

$$T(s) = c\deg(v_j) + c\log(s) + T(s-1) \qquad T(0) = c$$

Without loss of generality assume we relabel the vertices so that they are removed from Q in the order $u_n, u_{n-1}, \ldots, u_1$

So,

$$T(s) = \deg(u_s) + c\log(s) + T(s-1)$$

$$= \deg(u_s) + c\log(s) + \deg(u_{s-1}) + c\log(s-1) + T(s-2)$$

In general,

$$T(s) = \sum_{i=0}^{k-1} \deg(u_{s-i}) + \sum_{i=0}^{k-1} c\log(s-i) + T(s-k)$$

Choose k s.t.

$s - k = 0$

$k = s$

$$T(s) = \sum_{i=0}^{s-1} \deg(u_{s-i}) + \sum_{i=0}^{s-1} c\log(s-i) + c$$

Note: Starting size $= n$

$$T(n) = \underbrace{\sum_{i=0}^{n-1} \deg(v_i)}_{2m} + \underbrace{\sum_{i=0}^{n-1} c\log(n-i)}_{\simeq n\log(n)} + c$$

Thus, $T(n) \in \Theta(m + n\log(n))$